NUSANTARA BANK INDONESIA • ENTERPRISE DATA GOVERNANCE

Semantic Dictionary & Business Glossary

Self-Service Banking Operations: ATM & CRM Infrastructure (v2.4)

This business semantic dictionary establishes standardized definitions, data types, and DAX calculation logic for Nusantara Bank's regional self-service terminal network. Designed to bridge executive reporting with data engineering, this document ensures consistent metric interpretation across regional operations (Kanwil), vendor management, and financial auditing teams.

1. CORE ENTITIES & DIMENSIONAL ATTRIBUTES

ATTRIBUTE NAME	DATA TYPE	BUSINESS DEFINITION & GOVERNANCE RULE
Regional Office (Kanwil)	`Text (String)`	Regional administrative headquarters (*Kantor Wilayah*) supervising terminal deployment, cash logistics, and vendor maintenance within a designated territory in Indonesia.
Address	`Text (String)`	Public commercial landmark address mapped to ensure precise Azure Maps geocoding while masking sensitive physical bank branch coordinates.
Vendor Name	`Text (String)`	Anonymized third-party hardware provider (e.g., *Java Systems Corp*, *Nusantara Tech Solutions*) responsible for terminal hardware supply and SLA compliance.

2. BASE OPERATIONAL METRICS

METRIC NAME	DAX FORMULATION	BUSINESS CONTEXT
Total ATMs	`Total ATMs = SUM('Regional Breakdown'[Total ATMs])`	Count of active cash-dispensing Automated Teller Machines deployed across the region.
Total CRMs	`Total CRMs = SUM('Regional Breakdown'[Total CRMs])`	Count of active Cash Recycling Machines supporting both cash deposits and withdrawals.
Total Machines	`Total Machines = [Total ATMs] + [Total CRMs]`	Combined volume of self-service banking terminals in service.

> **Data Governance Note:** All numeric volume metrics in this semantic layer reflect scaled operational figures calibrated by a 1.15 multiplier to preserve confidentiality while maintaining exact regional proportion and ratio relationships.

3. ADVANCED PERFORMANCE & RISK KPIS

KPI NAME	DAX FORMULATION	ANALYTICAL INTERPRETATION
ATM Ratio %	`ATM Ratio % = DIVIDE([Total ATMs], [Total Machines], 0)`	Proportion of legacy cash dispensers within the regional machine fleet. Formatted as Percentage (1 decimal).
CRM Ratio %	`CRM Ratio % = DIVIDE([Total CRMs], [Total Machines], 0)`	Modernization indicator measuring recycling terminal penetration. Formatted as Percentage (1 decimal).
Warranty Risk Rate %	`Warranty Risk Rate % = DIVIDE([High Risk (Out of Warranty)], [Total Machines], 0)`	Risk exposure metric quantifying hardware operating past vendor maintenance warranty coverage.

4. MODEL ARCHITECTURE & DAX GOVERNANCE

The measures defined in this glossary support Nusantara Bank's operational analytics platform. By standardizing the DAX calculations across both *ATM & CRM Regional Operations* and *Provider Risk* pages, the semantic layer eliminates discrepancies in ratio evaluations caused by direct column aggregations.

Key governance principles applied:

- **Strict Division Guardrails:** All ratio measures utilize explicit `DIVIDE()` logic to handle zero-denominator scenarios gracefully without returning division-by-zero errors.
- **Unidirectional Relationships:** Filter propagation flows strictly from single-key dimension tables (`Regional Office` and `Vendor Risk`) to fact tables, preserving engine throughput.
- **Standardized Formatting:** Percentages are formatted to one decimal place (`0.0%`), and volumes use standard integer thousand separators.

> **Audit Compliance:** Any modification to DAX formulas or measure definitions requires formal change approval from the Business Intelligence Steering Committee to ensure ongoing alignment with enterprise reporting standards.